

Mastering Quadratic Equations ✨

$\times^2$

$+$

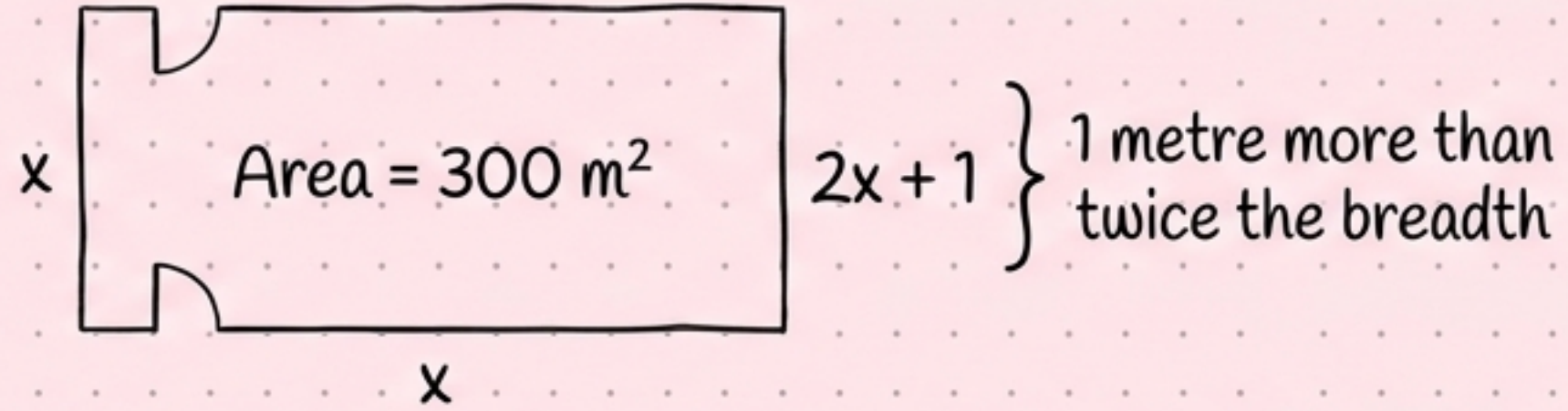


$+$

$\sqrt{\quad}$

Your ultimate visual cheat sheet for Chapter 4.
Swipe to unlock the math! ➡

Why do we need them?



$$\text{Area} = \text{Length} \times \text{Breadth} \Rightarrow x(2x + 1) = 300 \Rightarrow 2x^2 + x - 300 = 0$$

A 4,000-Year-Old Cheat Code



Babylonian

First to solve by finding two numbers with a given sum/product



Brahmagupta
C.E. 598–665

Gave an explicit formula



Al-Khwarizmi
C.E. 800

Studied different types of quadratics

The Anatomy of a Quadratic

$$ax^2 + bx + c = 0$$

The Real
Number Bosses
(Coefficients).

$$a \neq 0$$



The Degree 2
VIP!
(If this is
missing, it's not
quadratic).

Imposter Syndrome



Looks
cubic...

$$(x+2)^3 = x^3 - 4$$

↪ Simplifies to
 $6x^2 + 12x + 12 = 0$

It's Quadratic! ✓

Looks
quadratic...

$$x(x+1) + 8 = (x+2)(x-2)$$

↪ x^2 cancels out!
↪ Simplifies to
 $x + 12 = 0$

~~It's Linear! ✗~~

Rule: Always simplify before you judge!

The Factorisation Cheat Code



$$2x^2 - 5x + 3 = 0$$

Step 1: Multiply $a \times c$

$$2 \times 3 = 6$$

Step 2: Find factors of 6 that add to b (which is -5)

-2 and -3



Step 3: Split, Group & Extract!

$$\begin{aligned} 2x^2 - 2x - 3x + 3 = 0 &\rightarrow 2x(x - 1) - 3(x - 1) = 0 \\ &\rightarrow (2x - 3)(x - 1) = 0 \end{aligned}$$

Roots:
 $x = \frac{3}{2}$ and $x = 1$ 

The Magic Formula ✨

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



The Discriminant
(D) = $b^2 - 4ac$

This is the Heart of the Equation! It determines the **nature** of your roots without having to solve everything.

Nature of Roots: The Ultimate Matrix 🚦

● Green Light
($D > 0$)

Condition:
 $b^2 - 4ac > 0$

Meaning: Two distinct
real roots.



● Yellow Light
($D = 0$)

Condition:
 $b^2 - 4ac = 0$

Meaning: Two equal
real roots.



● Red Light
($D < 0$)

Condition:
 $b^2 - 4ac < 0$

Meaning: No real
roots.





Save this post for your exams!



Math doesn't have to be scary. Drop a  in the comments if this cheat sheet helped you, and tag a friend who needs it!